Riemann Surface

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§1 Riemann Surface

§1.1 Definition

A Riemann Surface is a topological space X equipped with a complex structure. A complex structure is a collection of local coordinates called **complex charts** on X

Definition. A **complex chart** on X is a homeomorphism $\phi : U \to V$ from an open set $U \subset X$ to an open set $V \subset \mathbb{C}$. A chart **centered at** p if $\phi(p) = 0$

Example 1.1

Let $X = \mathbb{R}^2$, the map

$$\phi : (x, y) \mapsto \frac{x + iy}{1 + \sqrt{x^2 + y^2}} = \frac{z}{1 + |z|}$$

ϕ a homeomorphism between $\mathbb{C}$ and $\mathbb{D}$, with continuous inverse

$$\phi^{-1} : (u, v) \mapsto \frac{u + iv}{1 - \sqrt{u^2 + v^2}} = \frac{w}{1 - |w|}$$

For any open set $U \subset \mathbb{R}^2$, ϕ_U is then a chart on $\mathbb{R}^2$

Example 1.2

For any chart $\phi : U \to V$ and an open subset $A \subset U$, $\phi_A : A \to \phi(U)$ is a chart on X called a **sub-chart** of ϕ.

The following practical condition is needed to make concepts as holomorphic maps between two Riemann surfaces well-defined

Definition. Two charts $\phi_1 : U_1 \to V_1$ and $\phi_2 : U_2 \to V_2$ are **compatible** if the transition map

$$T : \phi_2 \circ \phi_1^{-1} : \phi_1(U_1 \cap U_2) \to \phi_2(U_1 \cap U_2)$$

is holomorphic

> **Remark.** By proposition **??**, if $\phi_2 \circ \phi_1^{-1}$ is holomorphic, then so as $\phi_1 \circ \phi_2^{-1}$.

> **Remark.** Transition maps are conformal: differentiate $T \circ T^{-1} = id$

We want a collection of these charts which covers X, making sure each $p \in X$ have a chart containing p.

Definition. An **atlas** $\mathcal{A}$ on X is a collection $\mathcal{A} = \{\phi_\alpha : U_\alpha \to V_\alpha\}_{\alpha \in I}$ such that $X \subset \bigcup_{\alpha \in I} U_\alpha$ and $\phi_{\alpha_1}, \phi_{\alpha_2}$ are compatible for any $\alpha_1, \alpha_2 \in I$

We say two atlases $\mathcal{A}_1, \mathcal{A}_2$ on X are **equivalent** if every chart in $\mathcal{A}_1$ is compatible with every chart in $\mathcal{A}_2$, i.e., $\mathcal{A}_1 \cup \mathcal{A}_2$ is an atlas. Given an atlas $\mathcal{A}$, there exists a **maximal atlas** $\mathcal{F} = \bigcup_{\alpha \in I} \mathcal{A}_\alpha$, where each $\mathcal{A}_\alpha$ is an atlas compatible with $\mathcal{A}$.

Definition. A **complex structure** on X is a maximal atlas on X.

Definition (Riemann Surface)**.** A **Riemann Surface** is a second contable, Hausdorff, connected topological space X with a complex structure.

Remark. Some comments on structures of Riemann Surfaces:

1. A Riemann Surface X is a C^∞ 2-manifold: compose charts ϕ with the identification map $i : \mathbb{C} \to \mathbb{R}^2$ gives $\mathbb{R}^2$ charts and atlases. The transition maps are C^∞ because transition maps on X is holomorphic.
2. Orientation: if we define local orientation at $p \in X$ by a pullback to any chart containing p, the conformality of transition maps ensures that orientations are well-defined.
3. Connectedness and Path-Connectedness are equivalent
4. Classification of Compact Riemann Surfaces: by the previous comments and classification of orientable 2-dim C^∞ manifolds, every compact Riemann Surface is homeomorphic to a torus of genus g, for some $g \geq 0$.

Example 1.3 (Riemann Sphere)
Let $X = S^2 \subset \mathbb{R}^3$ and identify the plane $z = 0$ with $\mathbb{C}$. The atlas $\mathcal{A}$ consists of two charts:

1. Stereographic Projection from the north pole: $U_1 = S^1 - \{(0,0,1)\}, V_1 = \mathbb{C} - \{0\}$.

$$\phi_1(x,y,w) = \frac{x+iy}{1-w}$$

2. Stereographic projection from the south pole, followed by a conjugation: $U_2 = S^1 - \{(0,0,1)\}, V_2 = \mathbb{C} - \{0\}$.

$$\phi_2(x,y,w) = \frac{x-iy}{1+w}$$

The inverses are given by

$$z = u+iv, \phi_1^{-1}(z) = \left(\frac{2u}{|z|^2+1}, \frac{2v}{|z|^2+1}, \frac{|z|^2-1}{|z|^2+1}\right), \phi_2^{-1}(z) = \left(\frac{2u}{|z|^2+1}, \frac{-2v}{|z|^2+1}, \frac{1-|z|^2}{|z|^2+1}\right),$$

Compute that

$$\phi_2 \circ \phi_1^{-1}(z) = \phi_2\left(\frac{2u}{|z|^2+1}, \frac{2v}{|z|^2+1}, \frac{|z|^2-1}{|z|^2+1}\right) = \frac{2(u-iv)}{|z|^2+1}\frac{|z|^2+1}{2|z|^2} = \frac{z}{|z|^2} = \frac{1}{z}$$

which is holomorphic on $\mathbb{C} - \{0\}$. So the charts are compatible. $U_1 \cup U_2$ clearly covers S^2.$\mathcal{A}$ is an atlas on X.

§1.2 Examples of Riemann Surfaces

In this section we first outline a recipe for constructing Riemann Surfaces, use it to construct the complex projective line $\mathbb{CP}^1$, and present some other examples: the complex torus and smooth irreducible affine plane curves.

Motivation Given a topological space X and an open cover $\{U_\alpha\}_{\alpha \in I}$, a subset $U \subset X$ is open if and only if $U \cap U_\alpha$ is open in the subspace topology of U_α for all $\alpha \in I$.

Proof. If U is open, then $U \cap U_\alpha$ is open in U_α by the defn of the subspace topology. Conversely, if each $U \cap U_\alpha$ is open in U_α, then there exist open sets $V_\alpha \subset X$ such that $U \cap U_\alpha = V_\alpha \cap U_\alpha$. Since $\{U_\alpha\}$ covers X,

$$U = \bigcup_{\alpha \in I}(U \cap U_\alpha) = \bigcup_{\alpha \in I}(V_\alpha \cap U_\alpha),$$

a union of open sets, hence open. □

We could turn this observation around and build a recipe to define Riemann Surfaces by building an atlas and topology on a set X at the same time:

Recipe : Given the following data (no topology yet)

1. A set X and a countable collection of $\{U_\alpha\}_{\alpha\in I} \subset 2^X$ covering X.
2. Bijective mappings $\phi_\alpha : U_\alpha \to V_\alpha$ where V_α is an open subset of $\mathbb{C}$, such that the transition functions

$$\phi_\beta \circ \phi_\alpha^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \to \phi_\beta(U_\alpha \cap U_\beta)$$

   are holomorphic for all $\alpha, \beta \in I$.

We could make x into a topological space by:

1. Defining topology on each U_α: for each $\alpha \in I$, declare $U \subset U_\alpha$ to be open iff $\phi_\alpha(U)$ is an open subset of V_α (equivalently, $\mathbb{C}$)
2. Define topology on X: use the covering $\{U_\alpha\}$

$$\tau = \{U \subset X : U \cap U_\alpha \text{ is open in } U_\alpha \text{ for each } \alpha \in I\}$$

3. Check (X, τ) is connected + Hausdorff

The topological space (X, τ), along with the complex structure $\{(U_\alpha, \phi_\alpha)\}_{\alpha\in I}$, is a Riemann Surface.

We show the utility of this recipe by building the Riemann Surfaces $\mathbb{CP}^1$

Example 1.4 ($\mathbb{CP}^1$)

Let $\mathbb{CP}^1$ be the set of one dimensional subspaces of $\mathbb{C}^2$. Denote $[z : w] = \text{sp}(z, w)$. We use the following two sets to cover $\mathbb{CP}^1$:

$$U_1 = \{[z : w] | z \neq 0\}, U_2\{[z : w] | w \neq 0\},$$

The mappings sending lines to their slopes $\phi_1 : [z, w] \mapsto w/z, \phi_2 : [z, w] \mapsto z/w$ are bijective mappings $U_i \to \mathbb{C}^*$. To use the recipe it remains to check

Checklist:

1. Compatibility: we NTS $\phi_2 \circ \phi_1^{-1} : \phi_1(U_1 \cap U_2) \to \phi_2(U_1 \cap U_2)$ is holomorphic.

$$\phi_2 \circ \phi_1^{-1}(z) = \phi_2([1 : z]) = 1/z$$

   which is holomorphic on any open set not containing $\{0\}$: $\phi_i(U_1 \cap U_2) = \mathbb{C}^*$ is clearly open.

2. Connectedness: the homeomorphisms $\phi_i : U_i \to \mathbb{C}^*$ guarantees that U_i are connected open sets. As $X = U_1 \cup U_2$ and U_1, U_2 are non-disjoint it's not possible to disconnect X with disjoint open sets. X is connected.
3. Hausdorff: If two points p_1, p_2 are both in U_1, we can separate them by open sets: find disks $D_1, D_2 \subset \mathbb{C}$ containing $\phi_1(p_1)$ and $\phi_1(p_2)$, and then $\phi_1^{-1}(D_1), \phi_1^{-1}(D_2)$ separates p_1, p_2. Same goes with $p_1, p_2 \in U_2$.

   Lastly, if $p_1 \in U_1 - U_2, p_2 \in U_2 - U_1$, then by defn $p_1 = [1 : 0], p_2 = [0 : 1]$ and these are separable by $p_1 \in \phi_1^{-1}(D_1(0)), p_2 \in \phi_2^{-1}(D_1(0))$.The definition of ϕ_1, ϕ_2 ensures these are disjoint sets.

TODO: Is this the same as the quotient topology?

Example 1.5 (The complex torus $\mathbb{C}/L$)

Pick $\mathbb{R}$-linearly independent complex numbers ω_1, ω_2 and form the lattice

$$L = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 = \{a\omega_1 + b\omega_2 : a, b \in \mathbb{R}\}$$

Equip the quotient group $\mathbb{C}/L$ with quotient topology. Recall the quotient topology is the finest topology such that

$$\pi : \mathbb{C} \to \mathbb{C}/L, z \to z + L \text{ is continuous}$$

Explicitly,

$$U \subset \mathbb{C}/L \text{ is open } \iff \pi^{-1}(U) \text{ is open in } \mathbb{C}$$

When dealing with quotient spaces, it's usually easier to deal with the space we start with ($\mathbb{C}$ here) and pass to the quotient space ($\mathbb{C}/L$) via projection π.

Geometrically, quotient is another word for folding. The natural covering of $\mathbb{C}$ is just a bunch of disks centered at each point, we would like to cover $\mathbb{C}/L$ in a similar fashion: for each $z_0 \in \mathbb{C}$ we choose some disk D_{z_0} and hope $\{\pi_{z_0} : D_{z_0} \to \pi(D_{z_0})\}$ forms an atlas.

First we check that π is an open map. If $U \subset \mathbb{C}$ is open, then

$$\pi(U) = \bigcup_{u \in U} (u + L) \implies \pi^{-1}(\pi(U)) = U + L = \bigcup_{l \in L} (l + U)$$

which is a union of translation of open sets, hence open. So $\pi(U)$ is open.

We now make sure π is

1. Onto: By defn
2. Injective: Need to make the radius ϵ of D_{z_0} sufficiently small such that no two elements of D_{z_0} differs by a nonzero element of L.
3. Continuous: π is continuous, π_D is hence continuous with respect to subspace topology

π is now a homeomorphism! The atlas we want is $\{\pi_{z_0}^{-1} : \pi(D_{z_0}) \to D_{z_0}\}$. Let's check compatibility: let $U = \pi_{z_0}(D_{z_0} \cap D_{z_1}), V = \pi_{z_1}(D_{z_0} \cap D_{z_1})$.

$$T = \pi_1^{-1} \circ \pi_0 : U \to V$$

$T(z) = \pi_1^{-1}(z + L) = \implies \pi_1(T(z)) = z + L \implies T(z) = w(z) + z$. Note $w(z) = T(z) - z$ is a continuous function taking values in a discrete set L. We conclude w is locally constant, hence holomorphic.

§1.3 Affine Plane Curves

We now investigate a class of important Riemann Surfaces in Algebraic Geometry: zero locus of polynomials.

Example 1.6 (Baby Case: Graphs of Holomorphic Functions)

Let $U \subset \mathbb{C}$ be connected and $g_i : U \to \mathbb{C}$ holomorphic $(1 \leq i \leq n)$. Let $X = \{(z, g_1(z), \ldots, g_n(z)) | z \in U\}$ with subspace topology from $\mathbb{C}^{n+1}$ and define $\pi : X \to U$ by

$$(z, g_1(z), \ldots, g_n(z)) \mapsto z$$

Note $\pi^{-1} : z \to (z, g(z))$ and both π, π^{-1} are continuous. So π is a homeomorphism. Transition maps are the identity. Connectedness of X follows from assumption, and Hausdorff + 2nd Countable follows from $X \subset \mathbb{C}^{n+1}$ being a subspace.

$$X \text{ is a Riemann Surface with atlas } \{(X, \pi)\}$$

Definition. An **Affine plane curve** $X \subset \mathbb{C}^2$ is the zero locus of a polynomial $f(z, w)$. i.e. $X(f) = \{(z, w) : f(z, w) = 0\}$

To make the locus into a Riemann Surface like 1.6, we need a condition to make it locally a graph.

Definition. $f(z, w)$ is **non-singular** at p if eitehr $\partial f/\partial z(p) \neq 0$ or $\partial f/\partial w(p) \neq 0$. $X(f)$ is **non-singular** at $p \in X(f)$ if $f(z, w)$ is **non-singular**. Conveniently, we say $X(f)$ is **smooth** if it's non-singular at every point.

The first part of the definition is required to employ the **Implicit Function Theorem** to either w or z.

TODO: Prove this theorem in Appendix

Proposition 1.1 (Implicit Function Theorem)

Let $f(z, w) : \mathbb{C}^2 \to \mathbb{C}$, if there exists p such that $f(p) = 0$ and $\partial f/\partial w|_p \neq 0$, then there exist a unique holomorphic function $g : U \to V$ such that $f(z, g(z)) = 0$ for $z \in U$. Moreover, in U

$$g'(z) = -\frac{\partial f/\partial z}{\partial f/\partial w}$$

Corollary 1.1.1

If $h : W \to \mathbb{C}$ is holomorphic ($W \subset \mathbb{C}$ open), and $h'(w_0) \neq 0$, then ϕ is locally invertible near w_0.

Proof. Consider $f(z, w) = z - h(w)$. For $p = (h(w_0), w_0)$, $f(p) = 0$ and $\partial f(p)/\partial w = h'(w_0) \neq 0$. Hence by implicit function theorem there exist a neighborhood U of $h(w_0)$ and holomorphic function $g : U \to \mathbb{C}$ such that for all $z \in U$, $f(z, w) = 0$ iff $w = g(z)$ ($\iff z = h(g(z))$). So h is a left inverse of $g : U \to \mathbb{C}$, g is injective, and can be made surjective by mapping onto its image, hence $h : g(U) \to U$ is locally invertible.

Complex Structure of Affine Plane Curves: Let $X(f)$ be a smooth affine plane curve. For $p = (z, w) \in X$ such that $\partial f/\partial w|_p \neq 0$, there exist a neighborhood U of z, $g : U \to g(U)$ holomorphic, such that $f(z, w) = 0 \iff w = g(z)$ for $z \in U$. The charts on X are $(U \times g(U), \pi_z)$, where $\pi_z(z, w) = z$.

The transition maps are either the identity or a holomorphic function $g(z)$(depending on whether the maps are both π_z or π_w).

Finally, X is second countable and hausdorff as a subset of $\mathbb{C}^2$. However, connectedness is not granted:

$$f(z,w) = (z-w)(z-w-1)$$

The locus of zeroes is the union of two disjoint lines, hence disconnected. We cite the following theorem from TODO: cite lemma 7.3 of volume 2

Proposition 1.2

If $f(z,w) \in \mathbb{C}[z,w]$ is irreducible, then $X(f)$ is connected.

Therefore we reach at

Example 1.7

If $f(z,w) \in \mathbb{C}[z,w]$ irreducible and non-singular, then $X(f)$ is a Riemann Surface

Remark. No irreducible affine plane curve is compact, as for arbitrary $z_0 \in \mathbb{C}$, $f(z_0, w)$ always have a solution $w_0 \in \mathbb{C}$.

§1.4 Projective Curves

We remarked that compact Riemann surfaces can be classified up to homeomorphisms and that affine plane curves are non-compact. To make them compact and ask questions such as the genus of a curve, we can make them compact by projecting them.

Definition. The **complex projective plane** $\mathbb{CP}^n$ is defined as the set of one dimensional subspaces of $\mathbb{C}^{n+1}$. The **homogenous coordinates** of the line spanned by $(z_0, z_2, \cdots, z_n)$ is $[z_0 : z_1 : \cdots : z_n]$

We focus on the case $n = 2$.

Complex Structure on $\mathbb{CP}^2$: $\mathbb{CP}^2$ is the quotient of $\mathbb{C}^3$ by the action of multiplication of $\mathbb{C}^*$. So $\mathbb{CP}^2$ inherits a quotient topology from $\mathbb{C}^3$. Note for $U \subseteq \mathbb{C}^3$

$$\pi^{-1}(\pi(U)) = \bigcup_{\lambda \in C^*} \lambda U \subseteq \mathbb{C}^3 \implies \pi(U) \subseteq \mathbb{CP}^2$$

Hence π is an open map, by Prop **??** second countability follows. For an element in the complement of $R = \{(z,w) | z \sim w\}$, it lies in the preimage of C^* of the continuous function

$$f : \mathbb{C}^3 \to \mathbb{C}^3, (z,w) \mapsto z \wedge w$$

So $R \subsetneq X \times X$. By Prop **??**, $\mathbb{CP}^n$ is Hausdorff.

The atlas is

$$U_0 = \{[x:y:z] \in \mathbb{CP}^2 \mid x \neq 0\}, \qquad \varphi_0([x:y:z]) = \left(\frac{y}{x}, \frac{z}{x}\right) \tag{1}$$

$$U_1 = \{[x:y:z] \in \mathbb{CP}^2 \mid y \neq 0\}, \qquad \varphi_1([x:y:z]) = \left(\frac{z}{y}, \frac{x}{y}\right) \tag{2}$$

$$U_2 = \{[x:y:z] \in \mathbb{CP}^2 \mid z \neq 0\}, \qquad \varphi_2([x:y:z]) = \left(\frac{x}{z}, \frac{y}{z}\right) \tag{3}$$

$\varphi_i : U_i \to \mathbb{C}^2$ are clearly continuous maps onto their images, as $\widetilde{\varphi_0}$ is continuous.

$$\pi^{-1}(U_0) \xrightarrow{\pi} U_0 \xrightarrow{\varphi_0} \mathbb{C}^2 \setminus \{0\}, \quad \pi^{-1}(U_0) \xrightarrow{\widetilde{\varphi_0}} \mathbb{C}^2 \setminus \{0\}$$

Compatibility is easy.

Plane Curves on $\mathbb{CP}^2$: For a homogenous polynomial $F(x, y, z) \in \mathbb{C}[x, y, z]$, while it doesn't make sense to talk about evaluating F on $\mathbb{CP}^2$ (since $[x : y : z] = [\lambda x : \lambda y : \lambda z]$, $X(f)$ make sense in $\mathbb{CP}^2$.

Similar to affine plane curves, we would like to impose a non-singularity condition on F to make $X(f)$ a Riemann Surface.

Proposition 1.3 (Euler's Formula)
For a homogenous polynomial $F(x_1, \cdots, x_n)$

$$F(x_1, \cdots, x_n) = \frac{1}{d} \sum_{i=1}^{n} x_i \frac{\partial F}{\partial x_i}$$

where $d = \deg F$

Proof. prove for monomials, use linearity.

Definition. A homogenous polynomial $F(x, y, z)$ is **non-singular** if there's no solution to the equation in $\mathbb{CP}^2$ (equivalently, in $\mathbb{C} - \{0\}$)

$$F = \partial F/\partial x = \partial F/\partial y = \partial F/\partial z = 0$$

Lemma 1.1. Suppose that $F(x, y, z)$ is a homogeneous polynomial of degree d. Then F is nonsingular if and only if each X_i is a smooth affine plane curve in $\mathbb{C}^2$.

Proof. Suppose first that one of the X_i is not smooth; we may assume by symmetry that X_0 is not smooth. Define $f(u, v) = F(1, u, v)$, so that X_0 is defined by $f = 0$ in $\mathbb{C}^2$. Since X_0 is not smooth, there is a common solution $(u_0, v_0) \in \mathbb{C}^2$ to the set of equations

$$f = \frac{\partial f}{\partial u} = \frac{\partial f}{\partial v} = 0.$$

We claim then that $[1 : u_0 : v_0]$ is a common solution to the system (3.3), and thus F is singular. For this we note that

$$\begin{aligned}
F[1 : u_0 : v_0] &= f(u_0, v_0) = 0, \\
\frac{\partial F}{\partial y}[1 : u_0 : v_0] &= \frac{\partial f}{\partial u}(u_0, v_0) = 0, \\
\frac{\partial F}{\partial z}[1 : u_0 : v_0] &= \frac{\partial f}{\partial v}(u_0, v_0) = 0, \text{ and} \\
\frac{\partial F}{\partial x}[1 : u_0 : v_0] &= (dF - u_0 \frac{\partial F}{\partial y} - v_0 \frac{\partial F}{\partial z})[1 : u_0 : v_0] = 0,
\end{aligned}$$

where the last computation of $\partial F/\partial x$ uses Euler's formula (3.4). The proof of the other direction is very similar. □

Consider a nonsingular homogeneous polynomial $F(x, y, z)$ defining a projective plane curve $X \subset \mathbb{P}^2$. An important fact is that such nonsingular homogeneous polynomials are automatically irreducible. Each of the three affine pieces $X_i = X \cap \{x_i \neq 0\}$ then becomes a smooth irreducible affine plane curve, and by previous results, each is naturally a Riemann surface.

The coordinate charts on these affine pieces are given by ratios of coordinates. To verify that these locally defined complex structures are compatible, consider transition functions

between different affine patches. For instance, take a point $p \in X_0 \cap X_1$ where both x and y coordinates are nonzero. Near p, the curve X can be expressed as the graph of a holomorphic function h, so points have the form $[1 : w : h(w)]$. The transition from the chart $\phi_0 = y/x$ on X_0 to the chart $\phi_1 = z/y$ on X_1 is given by $w \mapsto h(w)/w$, which is holomorphic since $w \neq 0$ at p. Similar verifications for other chart combinations show that the complex structures glue together consistently.

Remark. Any projective plane curve is a closed subset of the compact space $\mathbb{P}^2$, hence a compact Riemann surface.

§1.5 Higher Dimensional Projective Spaces

We can also find Riemann surfaces embedded in higher-dimensional projective spaces.

Definition. The **projective space** $\mathbb{P}^n$ is defined as the vector space of 1-dimensional subspace of $\mathbb{C}^{n+1}$

The space $\mathbb{P}^n$ is covered by $n+1$ affine charts $U_i = \{[\mathbf{x}] : x_i \neq 0\}$, each isomorphic to $\mathbb{C}^n$ via the map that sends $[x_0 : \cdots : x_n]$ to $(x_0/x_i, \ldots, x_n/x_i)$ (omitting the i-th coordinate $x_i/x_i = 1$). These charts give $\mathbb{P}^n$ the structure of an n-dimensional complex manifold. The space is compact, as can be seen by projecting the unit sphere in $\mathbb{C}^{n+1}$ onto it.

If $F(x_0, \ldots, x_n)$ is a homogeneous polynomial, its zero locus in $\mathbb{P}^n$ is well-defined and is called a **hypersurface**.

Since $\mathbb{P}^n$ is an n-dimensional complex manifold, imposing one equation typically reduces the dimension by one. To obtain a Riemann surface (complex dimension 1) in $\mathbb{P}^n$, we consider the common zero locus of $n-1$ homogeneous polynomials.

Definition. Let $F_1, \ldots, F_{n-1}$ be homogeneous polynomials in variables $x_0, \ldots, x_n$, and let $X \subset \mathbb{P}^n$ be their common zero locus. We say X is a **smooth complete intersection curve** if the Jacobian matrix $(\partial F_i/\partial x_j)$ has maximal rank $n-1$ at every point of X.

Proposition 1.4

A smooth complete intersection curve in $\mathbb{P}^n$ is a compact Riemann surface. Local coordinates are given by ratios x_i/x_j of homogeneous coordinates.

The rank condition ensures via the Implicit Function Theorem that X is locally a graph of holomorphic functions.

Example 1.8

Not all projective curves are complete intersections. The **twisted cubic** $X \subset \mathbb{P}^3$, given by the image of $[x : y] \mapsto [x^3 : x^2y : xy^2 : y^3]$, is defined by three equations:

$$x_0x_3 = x_1x_2, \quad x_0x_2 = x_1^2, \quad x_1x_3 = x_2^2.$$

However, near any point, only two equations are needed to define X locally. For example, near $[1 : 0 : 0 : 0]$, the first two equations suffice to deduce the third

Definition. A **local complete intersection curve** in $\mathbb{P}^n$ is defined by homogeneous polynomials $\{F_\alpha\}$ such that near each point $p \in X$, the curve is described by $n-1$ polynomials $F_{\alpha_1}, \ldots, F_{\alpha_{n-1}}$ whose Jacobian has maximal rank $n-1$ at p.

Proposition 1.5

Every connected local complete intersection curve in $\mathbb{P}^n$ is a compact Riemann surface with local coordinates given by ratios of homogeneous coordinates.

A fundamental result states that every holomorphically embedded Riemann surface in projective space is a local complete intersection curve.

§1.6 Selected Solution to Exercises

§1.6.1 I.1

Ex (H)

Every atlas is equivalent to itself (definition of atlas); if $\mathcal{A}_1 \sim \mathcal{A}_2$, the inverse of transition maps are holomorphic, so $\mathcal{A}_2 \sim \mathcal{A}_1$; For $\mathcal{A}_1 \sim \mathcal{A}_2, \mathcal{A}_2 \sim \mathcal{A}_3$, we want to show

$$\phi_3 \circ \phi_1^{-1} : \phi_1(U_1 \cap U_3) \to \phi_3(U_1 \cap U_3)$$

is holomorphic. For each $p \in \phi_1(U_1 \cap U_3)$, there exist a chart domain U_2 of $\mathcal{A}_2$ which covers $\phi_1(p)$. Hence we can factor

$$\phi_3 \circ \phi_1^{-1} : \phi_1(U_1 \cap U_2 \cap U_3) \xrightarrow{\phi_2 \circ \phi_1^{-1}} \phi_2(U_1 \cap U_2 \cap U_3) \xrightarrow{\phi_3 \circ \phi_2^{-1}} \phi_3(U_1 \cap U_2 \cap U_3)$$

The factors are holomorphic maps restricted to an open subset of their domain, hence $\phi_3 \circ \phi_1$ is holomorphic.

§1.6.2 I.2

Ex (A)

Let τ_α be the topologies on

Ex (E)

Note this map is clearly well defined on $\mathbb{CP}^1$. $f^{-1} : S^2 \to \mathbb{CP}^1$ is given by

$$f^{-1}(a,b,c) \begin{cases} [0:1], \text{if } (a,b,c) = (0,0,1) \\ [1 : \frac{a+ib}{1-c}], \text{otherwise} \end{cases}$$

f is clearly continuous, $\mathbb{CP}^1$ is compact and S^2 is Hausdorff, hense by Prop **??**.

Ex (E)

Define $f : \mathbb{C}/L \to S^1 \times S^1$ by

$$a\omega_1 + b\omega_2 + L \mapsto (\exp(2\pi i a), \exp(2\pi i b))$$

well-definedness and bijectivity is evident. f is continuous as the lifting to $(f \circ \pi)(a\omega_1 + b\omega_2) = (\exp(2\pi i a), \exp(2\pi i b))$ is a continuous map $\mathbb{C} \to L$. Note f is a continuous bijection from a compact space to a Hausdorff space, so f is an open map by Prop **??**.

Ex (F)

For any $p = z + L \in \mathbb{C}/L$, clearly $q = z/n + L$ satisfies $n \cdot q = p$. If $z_1 + L, z_2 + L$ both satisfies $n \cdot (z_i + L) = p$, then

$$n \cdot (z_1 - z_2) + L = 0 + L$$

Hence counting total number of solutions amounts to counting distinct solutions to $n \cdot p = 0$ in $\mathbb{C}/L$. Write $p = a\omega_1 + b\omega_2$, plug in $n \cdot p = na\omega_1 + nb\omega_2 + L = 0 + L$, so $(na, nb) = (x, y)$ for some $(x, y) \in \mathbb{Z}^2$. There're exactly $|\{1, 2, \cdots, n\}^2| = n^2$ such pairs give rise to distinct solutions.

Ex (G)

Irreducibility: If $h(z) = g^2(z)$, then $f(z,w) = (w - g(z))(w + g(z))$. $f(z,w)$ isn't irreducible. Conversely, if $f(z,w) = h_1(z,w)h_2(z,w)$ where h_1, h_2 is non-constant, then

$$h_1(z,w)h_2(z,w) = w^2 - h(z)$$

WLOG $h(z) \neq 0$. We can write $h_1(z,w) = a_1 w + a_2 w^2 + \hat{h_1}(z,w)$ and $h_2(z,w) = b_1 w + b_2 w^2 + \hat{h_2}(z,w)$. Because the highest order of w is 2 and there're no cross terms in $w^2 - h(z)$, $\hat{h_1}, \hat{h_2}$ depends only on z and $a_1 = b_1 = 0$

$$h_1(z,w)h_2(z,w) = (a_1 w + \hat{h_1}(z))(b_1 w + \hat{h_2}(z))$$

Matching coefficients we find that $a_1 b_1 = 1$, hence we can assume that $a_1 = b_1 = 1$ (take a_1, b_1 out of the brackets and relabel $\hat{h_1}, \hat{h_2}$). Expanding and equating gives

$$w^2 + (\hat{h_1}(z) + \hat{h_2}(z))w + \hat{h_1}(z)\hat{h_2}(z) = w^2 - h(z) \implies \hat{h_1}(z) = -\hat{h_2}(z)$$

Finally $h(z) = w^2 - h_1(z,w)h_2(z,w) = w^2 - (w + \hat{h_1}(z))(w - \hat{h_1}(z)) = \hat{h_1^2}(z)$. So $h(z)$ is a perfect square.

Nonsingularity: Write $p = (z,w)$.. $\partial f/\partial w(p) = 2w \neq 0$ for $w \neq 0$. When $w = 0$, note any z such that $f(z,w) = 0$ is a root of $h(z)$. Hence $\partial f/\partial z(p) = h'(z) \neq 0$ for all such z if and only if $h(z)$ has no repetitive roots.

Ex (H)

Let $f(z,w) = az^2 + bzw + cw^2 + dz + ew + f$ affine plane conic singular at the point (z_0, w_0). ie

$$f(z_0, w_0) = 0 \tag{4}$$
$$\partial f/\partial z|_{(z_0,w_0)} = 2az_0 + bw_0 + d \tag{5}$$
$$\partial f/\partial w|_{(z_0,w_0)} = 2cw_0 + bz_0 + e \tag{6}$$

By a change of variables, we may assume that the singularity is at $(0,0)$:

$$g(z,w) = f(z + z_0, w + w_0) = f(z,w) - f$$

The last equality follows from a straightforward expansion subbing (2), (3).

$$g(z,w) = az^2 + bzw + cw^2 + dz + ew$$

As $\partial g/\partial z|_{(0,0)} = \partial g/\partial w|_{(0,0)} = 0 \implies d = e = 0$. It remains to show that $g(z,w)$ factors into linear polynomials.

Case 1: $b^2 - 4ac = 0$.
If $a = c = 0$ then $g(z,w) = bwz$. If $a \neq 0$ then $g(z,w) = a(z + b/2a)^2$. $c \neq 0$ is similar.

Case 2: $b^2 - 4ac \neq 0$.
The quadractic form

$$(x,x) = x^t \begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix} x$$

Is nondegenerate. There exist an orthogonal matrix P such that $P^t AP$ is diagonal. In otherwords there exist a linear change of variable $x = Py$ such that $(Py, Py) = \alpha\tilde{z}^2 - \gamma\tilde{w}^2$. This clearly can be factorzied in all cases.

TODO: Revise Quadratic Form theory

Ex

I. **Claim**: The locus of zeros of $f(z,w) = w^2 - (z^n - 1)$ is a smooth affine curve for $n \geq 1$.

As $h(z) = z^n - 1$ has distinct roots for $n \geq 1$, it's never a perfect square (the roots would need to have even multiplicity otherwise). By exercise I.2.G, $f(z,w)$ is non-singular and irreducible.

Ex (J)

For $\phi|_U$ to be a chart on $\mathbb{C}$, we only need it to be injective. Since surjectivity is automatic and holomorphic functions are open (so f^{-1}continuous). Consider the function $f(z,w) = w - \phi(z)$. As $f(p, g(p)) = 0$ and $\partial f/\partial z = \phi'(z) \neq 0$ at $(p, g(p))$, by the Implicit Function Theorem there exists a neighborhood U of p a unique holomorphic function (which is clearly ϕ) such that

$$f(z,w) = 0 \iff w = \phi(z)$$

§1.6.3 I.3

Ex (F)

$$F_0 = x_0x_3 - x_1x_2,\ F_1 = x_0^2 + x_1^2 + x_2^2 + x_3^2$$

Compute that

$$\left(\frac{\partial F_i}{\partial x_j}\right) = \begin{pmatrix} x_3 & -x_2 & -x_1 & x_0 \\ 2x_0 & 2x_1 & 2x_2 & 2x_3 \end{pmatrix}$$

Suppose that the two row vectors r_1, r_2 are linearly dependent $r_2 = \lambda r_1 (\lambda \in \mathbb{C})$ for some (x_0, x_1, x_2, x_3). Matching component $2, 3$ gives

$$\begin{cases} 2x_1 = -\lambda x_2 \\ 2x_2 = -\lambda x_1 \end{cases} \implies (\lambda^2 - 4)x_1 = (\lambda^2 - 4)x_2 = 0$$

Similarily the other columns gives $(\lambda^2 - 4)x_3 = (\lambda^2 - 4)x_0 = 0$. At least one of $x_i \neq 0$, so $\lambda \in \{\pm 2\}$. If $\lambda = 2$, then matching components gives $x_0 = x_3, x_1 = -x_2$. WLOG $x_0 \neq 0$, by linear dependence the det of all minors are zero, in particular

$$x_2x_3 + x_0x_1 = 0, x_1x_3 + x_0x_2 = 0$$

$$x_0(x_2 + x_1) = 0$$

§2 Morphisms between Riemann Surfaces

In this chapter, we look at morphisms between Riemann surfaces X and Y. We'll do this by first investigating the special case $Y = \mathbb{C}$, collect some interesting results and extend notions from complex analysis (holomorphic, meromorphic) there. We'll then move onto the global properties of holomorphic maps, starting with local normal form and notion of multiplicity and degree. This section ends with the Riemann Hurwitz formula, relating the genus of domain and codomain with information depending on the holomorphic map itself.

§2.1 Holomorphic and Meromorphic Functions

Let X be a Riemann surface and $f : X \to \mathbb{C}$, what does it mean for f to be holomorphic? We can bring the map (locally) to the level of $\mathbb{C}$:

Definition. Let $W \subset X$ be open. We say that $f : W \to \mathbb{C}$ is holomorphic at $p \in W$ if for every chart (U, ϕ) containing p, $f \circ \phi^{-1}$ is holomorphic at $\phi(p)$.

Example 2.1 (Chart Map)
Any complex chart $\phi : U \to V \subseteq \mathbb{C}$, considered as a function $\phi : U \to \mathbb{C}$, is holomorphic on its domain U.

Example 2.2 (Consistency with usual holomorphic maps $\mathbb{C} \to \mathbb{C}$)
Let $f : U \to \mathbb{C}$ be a function on an open set $U \subseteq \mathbb{C}$. The definition of holomorphicity on Riemann surfaces (considering $\mathbb{C}$ as a Riemann surface with the identity chart) coincides with the classical definition: f is holomorphic if it is complex differentiable at every point of U.

Example 2.3
If f and g are holomorphic at $p \in X$, then:

- $f \pm g$ is holomorphic at p
- fg is holomorphic at p
- If $g(p) \neq 0$, then f/g is holomorphic at p

Example 2.4
A function f defined near ∞ on the Riemann sphere $\mathbb{C}_\infty$ is holomorphic at ∞ if and only if the function $z \mapsto f(1/z)$ is holomorphic at $z = 0$. For a rational function $f(z) = p(z)/q(z)$ where p and q are polynomials, f is holomorphic at ∞ if and only if $\deg(p) \leq \deg(q)$.

Example 2.5
On the projective line $\mathbb{P}^1$ with homogeneous coordinates $[z : w]$, let $p(z, w)$ and $q(z, w)$ be homogeneous polynomials of the same degree d. If $q(z_0, w_0) \neq 0$, then the function

$$f([z : w]) = \frac{p(z, w)}{q(z, w)}$$

is well-defined and holomorphic in a neighborhood of $[z_0 : w_0]$.

Example 2.6

On a complex torus $\mathbb{C}/L$ with quotient map $\pi : \mathbb{C} \to \mathbb{C}/L$, a function $f : W \to \mathbb{C}$ (where $W \subseteq \mathbb{C}/L$ is open) is holomorphic at $p \in W$ if and only if there exists a preimage $z \in \pi^{-1}(p)$ such that $f \circ \pi$ is holomorphic at z. Moreover, f is holomorphic on W if and only if $f \circ \pi$ is holomorphic on $\pi^{-1}(W)$.

Example 2.7

Let $X = \{(z, w) \in \mathbb{C}^2 : F(z, w) = 0\}$ be a smooth affine plane curve defined by a nonsingular polynomial F. Then:

- The coordinate projections $\pi_z(z, w) = z$ and $\pi_w(z, w) = w$ are holomorphic functions on X
- Any polynomial function $g(z, w)$ restricts to a holomorphic function on X

Example 2.8

Let $X = \{[x : y : z] \in \mathbb{P}^2 : F(x, y, z) = 0\}$ be a smooth projective plane curve. At a point $p = [x_0 : y_0 : z_0]$ with $x_0 \neq 0$, the ratios y/x and z/x are holomorphic functions. More generally, if $G(x, y, z)$ and $H(x, y, z)$ are homogeneous polynomials of the same degree d and $H(x_0, y_0, z_0) \neq 0$, then the function

$$f([x : y : z]) = \frac{G(x, y, z)}{H(x, y, z)}$$

is holomorphic at p.

Let X be a Riemann surface and $p \in X$. For a function f holomorphic on a punctured neighborhood $U - \{p\}$, we classify its singularity at p as follows:

Definition. Let f be holomorphic on $U - \{p\}$.

- f has a **removable singularity** at p if there exists a chart ϕ with $p \in U$ such that $f \circ \phi^{-1}$ has a removable singularity at $\phi(p)$
- f has a **pole** at p if there exists a chart ϕ with $p \in U$ such that $f \circ \phi^{-1}$ has a pole at $\phi(p)$
- f has an **essential singularity** at p if there exists a chart ϕ with $p \in U$ such that $f \circ \phi^{-1}$ has an essential singularity at $\phi(p)$

Lemma 2.1. The type of singularity is chart-independent: f has a removable singularity (pole, essential singularity) at p if and only if for every chart ϕ with $p \in U$, $f \circ \phi^{-1}$ has the same type of singularity at $\phi(p)$.

The singularity type can be determined by the behavior of f near p:

- If $|f|$ is bounded near p, then f has a removable singularity
- If $|f| \to \infty$ as $x \to p$, then f has a pole
- If $|f|$ has no limit as $x \to p$, then f has an essential singularity

Definition. A function f is **meromorphic** at $p \in X$ if it is holomorphic, has a removable singularity, or has a pole at p. A function is meromorphic on $W \subseteq X$ if it is meromorphic at every point of W.

Example 2.9

On $\mathbb{C}$ (as a Riemann surface), the definition of meromorphic coincides with the classical notion.

Example 2.10

If f and g are meromorphic at p, then so are $f \pm g$, fg, and (if $g \not\equiv 0$) f/g.

Example 2.11

On the Riemann sphere $\mathbb{C}_\infty$, f is meromorphic at ∞ if and only if $f(1/z)$ is meromorphic at $z = 0$. All rational functions are meromorphic on $\mathbb{C}_\infty$.

Example 2.12

On $\mathbb{P}^1$ with homogeneous coordinates $[z : w]$, if $p(z, w)$ and $q(z, w)$ are homogeneous polynomials of the same degree with $q \not\equiv 0$, then $f([z : w]) = p(z, w)/q(z, w)$ is meromorphic on $\mathbb{P}^1$.

Example 2.13

On a complex torus $\mathbb{C}/L$ with projection $\pi : \mathbb{C} \to \mathbb{C}/L$, a function f is meromorphic at $p \in \mathbb{C}/L$ if and only if $f \circ \pi$ is meromorphic at some preimage of p. Elliptic functions correspond to meromorphic functions on $\mathbb{C}/L$.

Example 2.14

On a smooth projective plane curve $X = \{F(x, y, z) = 0\}$, if G and H are homogeneous polynomials of the same degree and H does not vanish identically on X, then G/H is meromorphic on X.

We now collect some of the generalizations of theorems from complex analysis

Theorem 2.2 (Identity Theorem)

Let X be a Riemann surface, $W \subset X$ connected and open, and f, g holomorphic functions on W. Suppose that $f = g$ on a subset $S \subset W$ which has a limit point in W, then $f = g$ on W.

Proof. It suffice to consider the case $g \equiv 0$.

$$U = \text{Int}(x \in X | f(x) = 0), V = U^c$$

U is open by definition and closed by continuity of $f \circ \phi^{-1}$ for any local chart ϕ. So U and V are clopen sets. U is non-empty by using the usual identity theorem locally (on a connected open neighborhood pf the limit point), hence $X = U$.

Theorem 2.3 (Discreteness of Zeros and Poles)

Let f be a meromorphic function defined on a connected open subset W of a Riemann surface X. If f is not identically zero, then the zeroes and poles of f form a discrete subset of W.

Proof. Let p be a zero of f and $\phi : U \to V$ a chart centered at p. $f \circ \phi^{-1}$ is meromorphic, hence has a discrete set of poles. The set of zeroes is also discrete by connectness of W and 2.2.

Corollary 2.0.1 (Finiteness of Zeroes and Poles on Compact Surfaces)

Any meromorphic function f on a compact Riemann surface which is not identically zero has a finite number of poles and zeros.

Proof. Follows from Theorem 2.3 and the fact that a discrete subset of compact space must be finite.

Theorem 2.4 (Maximum Modulus Principle)

Let $W \subset X$ be a connected open set. If there exist $p \in W$ such that $|f(x)| \leq |f(p)|$ for all $x \in X$, then $f(x) \equiv f(p)$.

Proof. f is locally a holomorphic function on a connected subset $U \subset \mathbb{C}$ which attains a maximum at p. By the usual maximum modulus principle $f = f(p)$ on U. By connectedness of W and theorem 2.2, $f = f(p)$ on W.

Lemma 2.1 (Constant Lemma)

Let X be a compact Riemann surface, the only holomorphic function $f : X \to \mathbb{C}$ are constant functions.

Proof. The map from $X \to \mathbb{C}$ by $x \mapsto |f(x)|$ is a continuous map from a compact space to $\mathbb{C}$, hence attains a maximum on X. X is connected, hence by the Maximum modulus principle f is a constant.

Theorem 2.5 (Open Mapping Theorem)

Any holomorphic function on a Riemann surface X is an open map.

§2.2 Examples of Meromorphic Functions

Due to Lemma 2.1, $O(X)$ is boring to study for compact Xs, but $M(X)$ remain lively. We now study the explicit forms of meromorphic functions on Riemann surfaces.

Riemann Sphere: we knew that rational functions $r(z) = p(z)/q(z)$ are meromorphic on $\mathbb{C}_\infty$. Actually, the converse is true as well:

Theorem 2.6

Any meromorphic function on $\mathbb{C}_\infty$ is a rational function.

Proof. Let $f : \mathbb{C}_\infty \to \mathbb{C}$ be meromorphic. As $\mathbb{C}_\infty$ is compact, by Corollary 2.0.1, there're finitely many distinct zeroes and poles $\{\lambda_i\}_{i=1}^n$ in $\mathbb{C}$. We assume $n \geq 1$ as otherwise f is holomorphic on $\mathbb{C}_\infty$, hence constant. Let $e_i = \mathrm{ord}_{\lambda_i}(f)$ and $r(z) = \prod_{i=1}^n (z - \lambda_i)^{e_i}$. The function $g(z) = f(z)/r(z)$ is holomorphic on $\mathbb{C}$, hence can be expanded as a Taylor series

$$g(z) = \sum_{n=0}^{\infty} c_n z^n$$

$r(z)$ is not constantly 0, hence $g(z) = f(z)/r(z)$ is meromorphic at ∞.

$$g(1/w) = \sum_{n=0}^{\infty} c_n w^{-n}$$

is meromorphic at 0 if and only if $c_n = 0$ for $n \geq 1$, so g is a constant and $f = gr$ is a rational function.

> **Remark.** This concludes that for any meromorphic function f on $\mathbb{C}_\infty$, $\sum_p \mathrm{ord}_p(f) = 0$ (sum over zeroes and poles).

Projective Line: The philosophy is to take a quotient of holomorphic functions on $\mathbb{C}^2$ which behaves nicely upon the action of $\mathbb{C}^*$. Their quotient descends to a well-defined meromorphic map on $(\mathbb{C}^2 - \{0\})/\mathbb{C}^*$.

Theorem 2.7

Any meromorphic function on $\mathbb{CP}^1$ is a quotient of homogenous polynomials of the same degree $f(z, w) = p(z, w)/q(z, w)$, where q is not identically 0.

Proof. The proof idea is the same as Theorem 2.6. Let f be a meromorphic function on $\mathbb{CP}^1$ By compactness of $\mathbb{CP}^1$ there're finitely many distinct zeroes and poles $\{[a_i : b_i]\}$, where $\mathrm{ord}_{[a_i,b_i]} f = e_i$.

Consider $r(z, w) = w^n \prod_i (b_i z - a_i w)^{e_i}$, where $n = -\sum_i e_i$. Let $g = f/r$, which has no zeroes and poles, except possibly at $[1 : 0]$.

If $[1 : 0]$ is not a pole of g, g is holomorphic on $\mathbb{CP}^1$, hence constant. f is a constant multiple of r and we're done. Otherwise, $1/g$ has no poles (holomorphic) and has a zero, so is constantly zero. This is impossible as $1/g = r/f$ isn't constant. □

Torus: the idea is to take quotients of functions on $\mathbb{C}$. Holomorphic functions on $\mathbb{C}/\Lambda$ is not an option as they're just constants. We could, however, mimic the case of projective line and choose functions which are well-behaved upon action by Λ.

The Theta function is defined as

$$\theta(z) = \sum_{n=-\infty}^{\infty} e^{\pi i[n^2\tau + 2nz]}$$

The action of Λ has the following effect

$$\theta(z + 1) = \theta(z) \tag{7}$$
$$\theta(z + \tau) = e^{-\pi i[\tau + 2z]}\theta(z) \tag{8}$$

The first is easy, the second follows from

$$\begin{aligned}\theta(z+\tau) &= \sum_{n=-\infty}^{\infty} e^{\pi i[n^2\tau+2n\tau+2nz]} \\ &= \sum_{n=-\infty}^{\infty} e^{\pi i[(n+1)^2\tau-\tau+2(n+1)z-2z]} \\ &= e^{-\pi i[\tau+2z]}\theta(z)\end{aligned}$$

We can verify that

$$\theta(1/2+\tau/2) = \sum_{n=-\infty}^{\infty} e^{\pi i[n^2\tau+n(1+\tau)]} = \sum_{n=-\infty}^{\infty} (-1)^n e^{\pi i n(n+1)}$$

Note $-\theta(1/2+\tau/2) = \sum_{n=-\infty}^{\infty}(-1)^{n-1}e^{\pi i n(n+1)} = \sum_{n=-\infty}^{\infty}(-1)^{n-1}e^{\pi i(n-1)n} = \theta(1/2+\tau/2)$ since $n^2 - n = n^2 + n \mod 2$. Hence $\theta(1/2+\tau/2)$. We now show this is in fact the unique zero of the theta function modular Λ.

Lemma 2.2 (Zeroes of Theta Functions)

$z_0 \in \mathbb{C}$ is a zero of $\theta(z)$ if and only if $z_0 + m + n\tau$ is for all $m, n \in \mathbb{Z}$, and orders of each zero are the same. The only zeroes are the simple zeroes $1/2 + \tau/2 + m + n\tau$.

Proof. The first statement is immediate from 7 and 8. For the second we use argument principle. A fundamental parallelogram D is a translation of the parallelogram with corners $0, 1, 1+\tau, \tau$. As the zeroes of θ are discrete, the number of zeroes in any compact set is finite, hence by induction we can find a fundamental parallelogram D, with corners $z_0, z_0 + 1, z_0 + 1 + \tau, z_0 + \tau$, and θ has no zeroes on ∂D.

Let $f(z) = \theta'(z)/\theta(z)$, differentiating 8 gives

$$\theta'(z+\tau) = -2\pi i\theta(z+\tau) + e^{-\pi i[\tau+2z]}\theta'(z) \implies f(z+\tau) = \frac{\theta'(z+\tau)}{\theta(z+\tau)} = -2\pi i + f(z)$$

Integrate f over d and use argument principle

$$\begin{aligned}\frac{1}{2\pi i}\int_{\partial D} f dz &= \sum_{i=1}^{4} \frac{1}{2\pi i}\int_{\gamma_i} f dz \\ &= \frac{1}{2\pi i}\left[\int_0^1 f(z_0+t)dt + \int_0^1 f(z_0+1+t\tau)dt - \int_0^1 f(z_0+\tau+(1-t))dt - \int_0^1 f(z_0+(1-t)\tau)dt\right] \\ &= \frac{1}{2\pi i}\left[\int_0^1 f(z_0+t)dt - \int_0^1 -2\pi i + f(z_0+(1-t))dt\right] \\ &= 1\end{aligned}$$

As θ have no poles, we conclude θ has 1 zero (counting multiplicity) inside D. □

We can translate the roots of these functions by defining

$$\theta^{(x)}(z) = \theta(z - 1/2 - \tau/2 - x)$$

So $\theta^{(x)}(z)$ has roots in $x + L$.

$$\theta^{(x)}(z+1) = \theta^{(x)}(z) \text{ and } \theta^{(x)}(z+\tau) = -e^{-2\pi i(z-x)}\theta^{(x)}(z)$$

Consider the ratio

$$R(z) = \frac{\prod_{i=1}^{m}\theta^{(x_i)}(z)}{\prod_{j=1}^{n}\theta^{(y_j)}(z)}$$

which is meromorphic on $\mathbb{C}$. Is $R(z)$ L periodic? $R(z+1) = R(z)$ is clear, and

$$R(z+\tau) = (-1)^{m-n} e^{2\pi i((m-n)z + \sum x_i - \sum y_j)} R(z)$$

Hence $R(z)$ is L-periodic if and only if $m = n$ and $\sum_i x_i - \sum_j y_j \in \mathbb{Z}$. Recall the correspondence between L-periodic meromorphic function on $\mathbb{C}$ and meromorphic functions on $\mathbb{C}/L$, we just proved

Proposition 2.1 (Ratios of Theta Family're Meromorphic on $\mathbb{C}/L$, $L = \mathbb{Z} + \mathbb{Z}\tau$)

For any positive integer d and two sets of d complex numbers $\{x_i\}$ and $\{y_i\}$ such that $\sum x_i - \sum y_j \in \mathbb{Z}$. The ratio

$$R(z) = \frac{\prod_{i=1}^{m} \theta^{(x_i)}(z)}{\prod_{j=1}^{n} \theta^{(y_j)}(z)}$$

is a meromorphic function on $\mathbb{C}/L$.

To fully prove that any meromorphic function on $\mathbb{C}/L$ is of this form requires a bit more degree theory, which we defer to 2.4.

Smooth Affine/Projective Plane Curves Let X be the affine plane curve defined by the zero locus of an irreducible nonsingular polynomial $f(z, w)$. Take any polynomials $g(z, w), h(z, w)$, g/h is meromorphic on X as long as h doesn't vanish on X.

So when does $X(f) \subset X(h)$? $f|h$ obviously would do, actually this is the only possibility given f is irreducible

Theorem 2.8 (Hillbert's Nullstellensatz)

Let f be an irreducible polynomial, any polynomial h vanishing where f does satisfies $f|h$.

Hence g/h is meromorphic on $X(f)$ if and only if $f \nmid h$.

The case for smooth projective plane curves is entirely similar. Let $F(x, y, z)$ be a irreducible non-singular polynomial, $X(F)$ be it's zero locus in $\mathbb{CP}^2$. For homogenous polynomials of the same degree $G(x, y, z), H(x, y, z)$, G/H is meromorphic on $X(F)$ given that H doesn't vanish on all of $X(F)$.

$$\begin{array}{ccc} F \text{ vanishes where } H \text{ does in } \mathbb{C}^3 & \Leftrightarrow & X(F) \subset X(H) \text{ (in } \mathbb{CP}^2) \\ \Updownarrow & & \Updownarrow \\ F(x,y,z) | H(x,y,z) & \Leftrightarrow & \left\{ \begin{array}{l} F(1,y,z) | H(1,y,z) \\ F(x,1,z) | H(x,1,z) \\ F(x,y,1) | H(x,y,1) \end{array} \right\} \end{array}$$

The equivalence on the left: by Nullstellensatz. The equivalence on the right: by applying Nullstellensatz argument on each X_i. This is a great example of using something geometric to prove something algebraic! (the bottom equivalence is nontrivial)

Smooth Projective Curves We would like to define curves in $\mathbb{CP}^n$

Definition. A subset $X \subset \mathbb{CP}^n$ is **holomorphically embedded** in $\mathbb{CP}^n$ if for each $p \in X$, there exist homoegenous coordinate z_j of p such that

1. $z_j \neq 0$
2. For each k, z_k/z_j is holomorphic near p.
3. There exist z_i such that z_i/z_j is a local coordinate on X near p.

A Holomorphically embedded Riemann Surface X is called a **smooth projective curve**.

TODO: condition 1 is obvious, 3 is the local complex analytic condition, why 2? Any counterexample?

Take the setup in the definition: at each $p \in X$, a ratio of homogenous coordinates $z_i/z_k = (z_i/z_j)/(z_k/z_j)$ would be meromorphic near p, as long as z_j isn't identically 0 on X. We can push that further by taking rational functions G/H of these ratios, where G, H are homogenous of the same degree, and homogenize by multiplying some power of z_j.

Proposition 2.2

For a smooth projective curve X, $G(z_0, \cdots, z_n), H(z_0, \cdots, z_n)$ homogenous and of the same degree. G/H is meromorphic near every point on X as long as H is not identically vanishing on X.

The objects in the projective space we investigated so far are all holomorphically embedded

Proposition 2.3

The projective line $\mathbb{CP}^1$, any smooth projective plane curve $X \subset \mathbb{CP}^2$, complete and local complete intersection curves, are all projective curves.

Proof. $[z_0 : z_1] \mapsto [z_0 : z_1 : 0 : \cdots : 0]$ maps $\mathbb{CP}^1$ into $\mathbb{CP}^n$. A local complete intersection curve looks like $[1 : g_1(w) : \cdots : g_n(w)]$ near any point p (or a permutation of that). So we can always choose a local coordinate such as $j = 0, i = 1$ which fits the definition, all ratios z_k/z_j is a holomorphic function near p.

§2.3 Holomorphic Maps $F : X \to Y$

Definition. Let X, Y be Riemann surfaces, a map $F : X \to Y$ is **holomorphic** at $p \in X$ if there exist charts $\phi_1 : U_1 \to V_1$ on X and $\phi_2 : U_2 \to V_2$ on Y such that

$$\phi_2 \circ F \circ \phi_1^{-1} : \phi_1(F^{-1}(U_2) \cap U_1) \to \phi_2(F(U_1) \cap U_2)$$

is holomorphic at $\phi_1(p)$. For $W \subset X$ open, F is holomorphic on W if it's holomorphic at all $p \in W$.

It's easy to check that holomorphicity is independent of the choice of charts using transition functions.

Lemma 2.3

Let $F : X \to Y$, $G : Y \to Z$ be holomorphic maps

1. F is continuous
2. $G \circ F : X \to Z$ is holomorphic.
3. Let $W \subset Y$ be open and $g : W \to Y$ a meromorphic, then $g \circ F$ is meromorphic on $F^{-1}(W)$, given that $F(X) \not\subset \{\text{poles of } g\}$

Proof. 1. Let $U \subset Y$ be open. For $p \in F^{-1}(U)$, pick charts $\phi_1 : U_1 \to V_1$ on X containing p and $\phi_2 : U_2 \to V_2$ containing $F(p)$. Note

$$F^{-1}(U_2 \cap U) = [\phi_1^{-1} \circ (\phi_2 \circ F \circ \phi_1^{-1})^{-1} \circ \phi_2](U_2 \cap U)$$

is open, as each of the map in the composition is open. Take a union over all such U_2s writes $F^{-1}(U)$ as a union of open sets, hence is open. F is continuous.

2. Let $p \in X$ and charts: $\phi_1 : U_1 \to V_1$ on X including p, $\phi_2 : U_2 \to V_2$ on Y including $F(p)$, and $\phi_3 : U_3 \to V_3$ on Z including $G \circ F(p)$. Note that

$$\phi_3 \circ (G \circ F) \circ \phi_1^{-1} = (\phi_3 \circ G \circ \phi_2^{-1}) \circ (\phi_2 \circ F \circ \phi_1^{-1})$$

F and G are holomorphic, by chain rule $G \circ F$ is holomorphic at p.

3. If g is a constant, then $g \circ F$ is in fact holomorphic. Otherwise the set of poles of g is discrete, by the identity theorem and $F(W) \not\subset \{\text{poles of } g\}$, $F^{-1}(\{\text{poles of g}\})$ is discrete on X, hence $g \circ F$ is meromorphic on $F^{-1}(W)$.

Definition. An **isomorphism** or **biholomorphism** $f : X \to Y$ is a bijective holomorphic map with holomorphic inverse. An **automorphism** on X is an isomorphism $f : X \to X$. The automorphisms on X is a group $Aut(X)$

We have shown in Problem I.2 (C) that $\mathbb{CP}^1 \cong \mathbb{C}_\infty$ via the isomorphism

$$[z : w] \mapsto (2\Re(z\bar{w}), 2\Im(z\bar{w}), |z|^2 - |w|^2)/(|z|^2 + |w|^2)$$

We now go over the theorems on holomorphic maps: most of them are really straight forward of employing known theorems from complex analysis locally, we will explain in length what that strategy means for 2.1 and outline ideas for the rest.

Proposition 2.4 (Open Mapping Theorem)
Any non-constant holomorphic map between Riemann Surfaces $F : X \to Y$ is open.

Proof. Let $U \subset X$ be open and $p \in U$. Pick charts $\phi_1 : U_1 \to V_1$ on X containing p and $\phi_2 : U_2 \to V_2$ on Y containing $F(p)$. $\phi_2 \circ f \circ \phi_1^{-1}$ is a non-constant holomorphic function as F is as well, hence is also open.

$$F(U \cap U_1) = \phi_2^{-1} \circ (\phi_2 \circ F \circ \phi_1^{-1}) \circ \phi_1(U \cap U_1)$$

is open as the composition are homeomorphisms/open maps. All such U_1 covers U, so $F(U)$ is open.

Proposition 2.5
A bijective holomorphic map $F : X \to Y$ is an isomorphism

Proof. This follows directly from using the corresponding theorem for holomorphic bijections $f : \mathbb{C} \to C$ locally.

Proposition 2.6 (Identity Theorem)
Let $F, G : X \to Y$ be holomorphic. If $F = G$ on a subset $S \subset X$ with a limit point in X, then $F = G$

Proof. If such a subset S and limit point $p \in X$ exist, choose charts as in proof of 2.1. $\phi_2 \circ F \circ \phi_1^{-1} = \phi_2 \circ G \circ \phi_1^{-1}$ on a subset of the chart domain with a limit point in it. Hence by the identity theorem on $\mathbb{C}$ $\phi_2 \circ F \circ \phi_1^{-1} = \phi_2 \circ G \circ \phi_1^{-1}$ and $F = G$ on an open subset of X. Now consider the clopen sets

$$U = \text{Int}(x \in X | F(x) = G(x)), V = U^c, X = U \cup V$$

By connectedness of X and $U \neq \emptyset$, $X = U$.

Proposition 2.7

Let X be a compact Riemann Surface, $F : X \to Y$ a non-constant holomorphic map. Then Y is compact and F is onto

Proof. $F(X)$ is open by the open mapping theorem and closed since $F(X)$ is a compact subset of the hausdorff space Y. The only non-empty clopen subset of a connected space is the space itself. $F(X) = Y$, and Y is compact.

Proposition 2.8 (Discreteness of Preimages)

Let $F : X \to Y$ be a non-constant holomorphic map. $|F^{-1}(y)|$ is discrete for all $y \in Y$ and is finite if X is compact.

Proof. If $S := F^{-1}(y)$ is non-discrete for some $y \in Y$, $F(X) = y$ by the identity theorem. If X, Y are compact, $F^{-1}(y) \neq \emptyset$ for all $y \in Y$ by prop 2.4, and is a finite set as it's a discrete subset of the compact space X.

Proposition 2.9 (Alternative definition of meromorphic functions)

For a meromorphic function $f : X \to \mathbb{C}$, define the map

$$F(x) = \begin{cases} f(x), & x \text{ is not a pole of } F \\ \infty, & x \text{ is a pole of } F \end{cases}$$

This is a 1-1 correspondence between

$$\{\text{Meromorphic Functions } f \text{ on } X\} \to \{\text{Holomorphic Maps } F : X \to \mathbb{C}_\infty, F \not\equiv \infty\}$$

We make one clarification: each Meromorphic Function f on X is formally a triplet (f, P) where $P \subset X$ is a discrete subset (representing poles), $f : X \backslash P \to \mathbb{C}$ is holomorphic.

Proof. For each meromorphic function f, the associated F is clearly holomorphic near every x which isn't a pole of f. If x is a pole of f, Let ϕ_1 be a chart on X centered at p and $\phi_2(z) = 1/z$ the chart on $\mathbb{C}_\infty$ near ∞.

$$\phi_2 \circ F \circ \phi_1^{-1}(z) = \frac{1}{F \circ \phi_1^{-1}(z)} \tag{9}$$

is bounded near 0 since $F \not\equiv \infty$, hence holomorphic at 0.

To check the correspondence is 1-1. It's clearly injective as where F isn't ∞ is entirely determined by $f(x)$. Now let $F : X \to \mathbb{C}_\infty$ be meromorphic and $F \not\equiv \infty$. Define $P = F^{-1}(\infty)$ and $f(x) = F(x)$ for $x \in X \backslash P$. f is holomorphic on $X \backslash P$ (since F is holomorphic on $\mathbb{C}$), and have poles at every point of $p \in P$ since (9) is holomorphic at 0.

An Application using this Correspondence:

§2.4 Global Property of Holomorphic Maps

It may be surprising, but every holomorphic map between Riemann Surfaces are locally a power map

Proposition 2.10 (Local Normal Form)

Let $F : X \to Y$ be a non-constant holomorphic map and $p \in X$. There exist an unique integer $m \geq 0$ with the following property: for every chart $\phi_2 : U_2 \to V_2$ on Y centered at $F(p)$, there exist a chart $\phi_1 : U_1 \to V_1$ centered at p such that $\phi_2 \circ F \circ \phi_1^{-1}(z) = z^m$

Proof. Let $\phi_2 : U_2 \to V_2$ be a chart centered at $F(p)$, and pick any chart $\psi : U \to V$ centered at p. $\phi_2 \circ F \circ \psi^{-1}(w)$ is holomorphic near 0 and can be expanded as $\sum_{n\geq 0} c_n w^n$. Let $m = \min\{n \in \mathbb{N} | c_n \neq 0\}$ and rewrite $\phi_2 \circ F \circ \psi^{-1}(w) = w^m h(w)$ for some h holomorphic and non-vanishing in a neighborhood of 0. Since h is holomorphic and $h(0) \neq 0$, there's an mth root $g(w)^m = h(z)$ near 0. Let $z = \eta(w) = wg(w)$, $\eta'(0) \neq 0$, hence by corollary 1.1.1, there exist a neighborhood V_1 of 0 where η is invertible. So $\phi_1 = \eta \circ \psi$ is a chart is centered at p and:

$$\begin{aligned}
\phi_2 \circ F \circ \phi_1^{-1}(z) &= \phi_2 \circ F \circ (\eta \circ \psi)^{-1}(z) \\
&= \phi_2 \circ F \circ \psi^{-1}(\eta^{-1}(z)) \\
&= \phi_2 \circ F \circ \psi^{-1}(w) \\
&= w^m h(w) \\
&= (wg(w))^m \\
&= z^m
\end{aligned}$$

For uniqueness, note that if F is locally z^m with respect to charts $\phi_1 : U_1 \to V_1, \phi_2 : U_2 \to V_2$.

$$\phi_2 \circ F \circ \phi_1^{-1}(z) : \phi_1(F^{-1}(U_2) \cap U_1) \to \phi_2(U_2 \cap F(U_1)), z \mapsto z^m$$

We note, by bijectiveity of ϕ_1 and ϕ_2, that $|F^{-1}(q)|$ (where F is restricted to $F^{-1}(U_2) \cap U_1$) for any $q \in U_2 \cap F(U_1)$ is precisely m!

Definition. The **multiplicity** of F at p, denoted $\text{mult}_p(F)$, is the unique integer m such that there're local coordinates at p and $F(p)$ with F locally being $z \mapsto z^m$.

Definition. Given a non-constant holomorphic function $F : X \to Y$ between compact Riemann surfaces. The degree of F is defined as

$$d_y(F) = \sum_{x \in F^{-1}(q)} \text{mult}_x(F)$$

for any $y \in Y$

Lemma 2.4

Degree of F is well defined (independent of $y \in Y$)

Proof. e

Meromorphic Functions on the Torus, Reprise: As an application of degree theory, we finish the complete description of $\mathcal{M}(\mathbb{C}/L)$ as promised.

Proposition 2.11

Any meromorphic function on the torus is a ratio of translated theta functions.

§2.5 Riemann Hurwitz Formula: Relating the Genus

Definition. Let S be a compact 2-manifold (possibly with boundary). A **triangulation** of S is a decomposition into closed subsets, each homeomorphic to a triangle, such that any two triangles intersect only at vertices or along entire edges. For a triangulation with v vertices, e edges, and t triangles, the **Euler number** is defined as:

$$e(S) = v - e + t$$

Proposition 2.12

The Euler number $e(S)$ is independent of the choice of triangulation. For a compact orientable 2-manifold without boundary of genus g, the Euler number is:

$$e(S) = 2 - 2g$$

Proof. The proof proceeds in three steps:

1. **Invariance under refinement**: Consider elementary refinements:

   - Adding a vertex in the interior of a triangle and connecting it to the three vertices: changes counts by $\Delta v = +1$, $\Delta e = +3$, $\Delta t = +2$, so $\Delta e(S) = 1 - 3 + 2 = 0$
   - Adding a vertex on an edge (shared by two triangles) and connecting to opposite vertices: changes counts by $\Delta v = +1$, $\Delta e = +3$, $\Delta t = +2$, so $\Delta e(S) = 0$
   - For surfaces with boundary, bisecting a boundary triangle: $\Delta v = +1$, $\Delta e = +2$, $\Delta t = +1$, so $\Delta e(S) = 0$

   Since all elementary refinements preserve $e(S)$, and any refinement is a sequence of elementary ones, $e(S)$ is refinement-invariant.

2. **Existence of common refinement**: Any two triangulations of a compact 2-manifold have a common refinement (obtained by superimposing them and adding necessary vertices and edges).

3. **Computation for standard surfaces**:

$$\begin{aligned} \text{Sphere: } & e(S^2) = 2 \\ \text{Cylinder: } & e(S^1 \times I) = 0 \\ \text{Disk: } & e(D^2) = 1 \end{aligned}$$

The genus g surface can be obtained from the sphere by attaching g handles. Each handle attachment decreases the Euler number by 2:

$$e(\Sigma_g) = e(S^2) - 2g = 2 - 2g$$

This follows from the formula for connected sum: $e(\Sigma_1 \# \Sigma_2) = e(\Sigma_1) + e(\Sigma_2) - 2$. □

Theorem 2.9 (Hurwitz's Formula)

Let $F : X \to Y$ be a nonconstant holomorphic map between compact Riemann surfaces. Then:

$$2g(X) - 2 = \deg(F)(2g(Y) - 2) + \sum_{p \in X} [\text{mult}_p(F) - 1]$$

where the sum is over all ramification points of F.

Proof. Choose a triangulation of Y such that all branch points of F are vertices. Let the triangulation have:

$$\begin{aligned} v &= \text{number of vertices} \\ e &= \text{number of edges} \\ t &= \text{number of triangles} \end{aligned}$$

Lift this triangulation to X via F^{-1}. Since F is a degree $d = \deg(F)$ covering map away from branch points, we have:

$$\begin{aligned} t' &= d \cdot t \\ e' &= d \cdot e \end{aligned}$$

For vertices, the counting is more subtle. For each vertex $q \in Y$, the number of preimages is:

$$|F^{-1}(q)| = \sum_{p \in F^{-1}(q)} 1 = d - \sum_{p \in F^{-1}(q)} (\text{mult}_p(F) - 1)$$

Therefore, the total number of vertices in X is:

$$v' = \sum_{q \in \text{vertices}(Y)} |F^{-1}(q)| = d \cdot v - \sum_{p \in \text{ramification points}} (\text{mult}_p(F) - 1)$$

Now compute the Euler numbers:

$$\begin{aligned} e(X) &= v' - e' + t' \\ &= \left[d \cdot v - \sum_p (\text{mult}_p(F) - 1) \right] - d \cdot e + d \cdot t \\ &= d(v - e + t) - \sum_p (\text{mult}_p(F) - 1) \\ &= d \cdot e(Y) - \sum_p (\text{mult}_p(F) - 1) \end{aligned}$$

Using $e(X) = 2 - 2g(X)$ and $e(Y) = 2 - 2g(Y)$, we obtain:

$$2 - 2g(X) = d(2 - 2g(Y)) - \sum_p (\text{mult}_p(F) - 1)$$

Rearranging gives Hurwitz's formula:

$$2g(X) - 2 = d(2g(Y) - 2) + \sum_p (\text{mult}_p(F) - 1)$$

□

The formula relates the genera of X and Y, the degree of the map, and the total ramification. It is fundamental in the study of holomorphic maps between Riemann surfaces.

§2.6 Selected Solutions to Exercises

§2.7 II.1

Ex (H)

Let $f : \mathbb{C} \to \mathbb{C}$ be bounded and entire. Then $g(z) = f(1/z) : \mathbb{C}^* \to \mathbb{C}$ is bounded near 0 and holomorphic on $\mathbb{C}^*$. By Riemann's removable singularity theorem, there exist $p \in \mathbb{C}$ such that the extension

$$g(z) = \begin{cases} f(1/z) & z \neq 0 \\ p & z = 0 \end{cases}$$

is holomorphic on $\mathbb{C}$. Hence $\hat{f} : \mathbb{C}_\infty \to \mathbb{C}$ by

$$\hat{f}(z) = \begin{cases} f(z) & z \in \mathbb{C} \\ p & z = \infty \end{cases}$$

is holomorphic on $\mathbb{C}_\infty$. $\mathbb{C}_\infty$ is compact, hence by theorem 1.38 $\hat{f}$ (hence f) is constant.

Ex (I)

One can just use theorem 1.38. Let's do this without invoking the theorem, let $f(z) = \frac{p(z)}{q(z)} = \frac{a_m z^m + \cdots + a_0}{b_n z^n + \cdots + b_0}$ be reduced (there isn't a common zero of p and q).

By f holomorphic at ∞, $f(1/z)$ is holomorphic at 0 (bounded near 0), which happens iff $m \leq n$. On the other hand, f is holomorphic at 0, so $f(z)$ is bounded near 0. This happens iff $m \geq n$, so $m = n$.

Any zero of q would be a pole of f, hence $n = 0$ and $m = n = 0 \implies f$ is constant.

§2.7.1 II.2

Ex (A)

Let $p = [z_0, w_0]$, choose $[z_1 : w_1] \in \mathbb{CP}^1 \backslash (S \cup \{[z_0 : w_0]\})$

$$f([z : w]) = \frac{z_0 w - w_0 z}{z_1 w - w_1 z}$$

Ex (G)

Let $R = I + I\tau$ where $I = (-1, 1)$. Since

$$\sum p_i = \sum q_i \iff \pi\left(\sum x_i - \sum y_i\right) = 0 \iff \sum x_i - \sum y_i \in L$$

To find $\{x_i\}, \{y_i\} \in \mathbb{C}$ given $\{p_i\}, \{q_i\} \subset \mathbb{C}/L$ as asked in the problem it suffice to find $\{x_i\}, \{y_i\}$ such that $\sum_i x_i - \sum_i y_i \in R$.

We induct on d. When $d = 1$ we can write $x_1 = a + b\tau, y_1 = c + d\tau$ for $a, b, c, d \in I$. Obviously one can always adjust the sign of c such that $a - c, b - d \in I$. For the inductive step, if we can find $\{x_i\}_{i=1}^d, \{y_i\}_{i=1}^d$ such that $\sum_{i=1}^d x_i - \sum_{i=1}^d y_i \in R$, we can certainly repeat the same procedure in the base case and adjust x_{d+1}, y_{d+1} accordingly.

§3 Integration on Riemann Surfaces

We explore the formal expression $f(z)dz$ and how to integrate it.

§3.1 Differential Forms

Definition. A **holomorphcic 1-form** on an open set $V \subset \mathbb{C}$ is a formal expression $\omega = f(z)dz$ where $f : V \to \mathbb{C}$ is holomorphic. We say that ω is a holomorphic 1-form in the coordinate z.

In the following text we'll just say 1-form instead of holomorphic 1-form.

Definition. Let $\omega_1 = f(z)dz, \omega_2 = g(w)dw$ be 1-forms on open subsets $V_1, V_2 \subset \mathbb{C}, T : V_2 \to V_1$ a holomorphic map. We say ω_1 transforms to ω_2 under $w = T(z)$ if $g(w) = f(T(w))T'(w)$

This is exactly what we want to do in calculus:

$$f(z)dz = f(T(w))T'(w)dw$$

The condition ensures the form we want to integrate to be unchanged upon change of coordinate

Definition. A **holomorphic 1-form** on X is a collection of 1-forms $\{\omega_\phi\}$, one for each chart $\phi : U \to V$ on X. Such that for overlapping charts $\phi_i : U_i \to V_i (i = 1, 2)$, ω_{ϕ_1} transforms to ω_{ϕ_2} under $T = \phi_1 \circ \phi_2^{-1}$.[1]

One might note that the definition requires assigning a holomorphic 1-form ω for all charts on X, not just those in a fixed atlas. The following lemma shows that it suffices to specify ωs on an atlas.

Lemma 3.1

Let $\mathcal{A}$ be a complex atlas on X. Suppose that 1-forms are given for each chart of $\mathcal{A}$, which transform to each other on their common domains. Then there exists a unique 1-form on X extending these holomorphic 1-forms on each of the charts of $\mathcal{A}$.

Proof. Fix a one form $\omega_\phi = f(z)dz$ on a chart $\phi : U \to V$ in $\mathcal{A}$. For any chart $\phi_1 : U_1 \to V_1$, not in $\mathcal{A}$ and overlapping with ϕ, define $\omega_{\phi_1} = f(T(w))T'(w)dw$ where $T = \phi_1 \circ \phi_2^{-1}$.

We first show that this is independent of the choice of charts in $\mathcal{A}$: suppose that we have overlapping charts ϕ, ψ in $\mathcal{A}$ and θ not in $\mathcal{A}$. Write $\omega_\phi = f(z)dz, \omega_\psi = g(w)dw, \omega_\theta = h(\xi)d\xi$, and name transition maps $z = T(\xi), w = S(\xi)$

Definition. Let ω be a meromorphic 1-form defined in a neighborhood of p, choose coordinate z centered at p and write $\omega = f(z)dz$ where f is meromorphic at $z = 0$. The **order** of ω at p, $\mathrm{ord}_p(\omega)$, is defined to be $\mathrm{ord}_0(f)$.

p is called a zero of ω of order n if $\mathrm{ord}_p(\omega) = n > 0$ and a pole of ω of order n if $\mathrm{ord}_p(\omega) = -n < 0$

If we choose a different coordinate w centered at p, and $\omega = g(w)dw$, then $g(w) = f(T(w))T'(w)$ near p, and obviously $\mathrm{ord}_0(f) = \mathrm{ord}_0(g)$.

Formalism: Using $z = x + iy$ and $\overline{z} = x - iy$, we have:

$$x = \frac{z + \overline{z}}{2}, \quad y = \frac{z - \overline{z}}{2i}$$

$$dx = \frac{dz + d\overline{z}}{2}, \quad dy = \frac{dz - d\overline{z}}{2i}$$

[1] We only require $g(w) = f(T(w))T'(w)$ on the domain of T

Define the differential operators:

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right), \quad \frac{\partial}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$$

A function f is holomorphic if and only if $\partial f/\partial \overline{z} = 0$.

The most basic form we'll be using is the following

Definition. A C^∞ 1-form on $V \subset \mathbb{C}$ is an expression:

$$\omega = f(z, \overline{z})dz + g(z, \overline{z})d\overline{z}$$

where f and g are C^∞ functions.

Under holomorphic coordinate change $z = T(w)$, the transformation rule is:

$$f_2(w) = f_1(T(w))T'(w), \quad g_2(w) = g_1(T(w))\overline{T'(w)}$$

Definition. A C^∞ 1-form on a Riemann surface X is a collection of local 1-forms that transform correctly under chart transitions.

Lemma 3.2

C^∞ 1-forms defined on an atlas that transform consistently extend uniquely to the entire surface.

Definition. A 1-form is of type $(1,0)$ if locally $\omega = f(z, \overline{z})dz$, and of type $(0,1)$ if locally $\omega = g(z, \overline{z})d\overline{z}$.

Holomorphic 1-forms are of type $(1,0)$. The $(1,0)$ and $(0,1)$ parts transform separately under holomorphic coordinate changes.

Definition. A C^∞ 2-form on $V \subset \mathbb{C}$ is:

$$\eta = f(z, \overline{z})dz \wedge d\overline{z}$$

with f C^∞. We have $dz \wedge dz = d\overline{z} \wedge d\overline{z} = 0$ and $dz \wedge d\overline{z} = -d\overline{z} \wedge dz$.

Under $z = T(w)$, the transformation is:

$$g(w) = f(T(w))|T'(w)|^2$$

Definition. A C^∞ 2-form on X is a collection of local 2-forms transforming correctly under chart transitions.

Lemma 3.3

C^∞ 2-forms defined consistently on an atlas extend uniquely to the entire surface.

§3.2 Operations on Forms

Operations on Differential Forms

This section develops the fundamental operations one can perform with differential forms on Riemann surfaces, including multiplication by functions, differentiation, wedge products, and pullbacks. These operations provide the essential toolkit for working with forms locally and globally.

Multiplication by Functions

Definition. Let h be a C^∞ function and ω a C^∞ 1-form on a Riemann surface X. Locally, if $\omega = f dz + g d\bar{z}$, define the product $h\omega$ by:

$$h\omega = (hf)dz + (hg)d\bar{z}$$

This gives a well-defined C^∞ 1-form on X.

Properties:

- If ω is of type $(1,0)$ or $(0,1)$, so is $h\omega$
- If ω and h are holomorphic, then $h\omega$ is holomorphic
- If ω and h are meromorphic, then $h\omega$ is meromorphic with:

$$\mathrm{ord}_p(h\omega) = \mathrm{ord}_p(h) + \mathrm{ord}_p(\omega)$$

Similarly, for a C^∞ 2-form $\eta = f dz \wedge d\bar{z}$, define:

$$h\eta = (hf)dz \wedge d\bar{z}$$

Differentials of Functions

Definition. For a C^∞ function f on X, define the 1-forms:

$$\begin{aligned} \partial f &= \frac{\partial f}{\partial z} dz \\ \bar{\partial} f &= \frac{\partial f}{\partial \bar{z}} d\bar{z} \\ df &= \partial f + \bar{\partial} f = \frac{\partial f}{\partial z} dz + \frac{\partial f}{\partial \bar{z}} d\bar{z} \end{aligned}$$

Lemma 3.4

The operators $d, \partial, \bar{\partial}$ are well-defined and $\mathbb{C}$-linear. They satisfy the product rule:

$$d(fg) = f dg + g df, \quad \partial(fg) = f\partial g + g\partial f, \quad \bar{\partial}(fg) = f\bar{\partial} g + g\bar{\partial} f$$

A function f is holomorphic if and only if $\bar{\partial} f = 0$.

A 1-form ω is **exact** on U if $\omega = df$ for some C^∞ function f on U.

Wedge Product of 1-Forms

Definition. For C^∞ 1-forms $\omega_1 = f_1 dz + g_1 d\bar{z}$ and $\omega_2 = f_2 dz + g_2 d\bar{z}$, define their wedge product as:

$$\omega_1 \wedge \omega_2 = (f_1 g_2 - f_2 g_1) dz \wedge d\bar{z}$$

This gives a well-defined C^∞ 2-form on X.

Differentiating 1-Forms

Definition. For a C^∞ 1-form $\omega = f dz + g d\bar{z}$, define:

$$\begin{aligned} \partial\omega &= \frac{\partial g}{\partial z} dz \wedge d\bar{z} \\ \bar{\partial}\omega &= -\frac{\partial f}{\partial \bar{z}} dz \wedge d\bar{z} \\ d\omega &= \partial\omega + \bar{\partial}\omega = \left(\frac{\partial g}{\partial z} - \frac{\partial f}{\partial \bar{z}} \right) dz \wedge d\bar{z} \end{aligned}$$

Lemma 3.5

These are well-defined C^∞ 2-forms. The operators satisfy:

$$d(f\omega) = df \wedge \omega + f d\omega$$
$$\partial(f\omega) = \partial f \wedge \omega + f\partial\omega$$
$$\bar{\partial}(f\omega) = \bar{\partial} f \wedge \omega + f\bar{\partial}\omega$$

and $ddf = \partial\partial f = \bar{\partial}\bar{\partial} f = 0$ for any C^∞ function f.

A function f is **harmonic** if $\partial\bar{\partial} f = 0$. A 1-form ω is:

- **closed** if $d\omega = 0$
- **∂-closed** if $\partial\omega = 0$
- **$\bar{\partial}$-closed** if $\bar{\partial}\omega = 0$

Lemma 3.6

A holomorphic 1-form is closed. Conversely, a closed $(1,0)$-form is holomorphic.

Pulling Back Differential Forms

Definition. Let $F : X \to Y$ be a holomorphic map between Riemann surfaces. For a C^∞ 1-form $\omega = f dz + g d\bar{z}$ on Y, define its pullback by:

$$F^*\omega = f(h(w), \overline{h(w)})h'(w)dw + g(h(w), \overline{h(w)})\overline{h'(w)}d\bar{w}$$

where $z = h(w)$ is the local expression of F.

Lemma 3.7

The pullback $F^*\omega$ is a well-defined C^∞ 1-form on X that preserves:

- Holomorphicity
- Meromorphicity
- Type $(1,0)$ or $(0,1)$

The pullback commutes with differentiation:

$$F^*(df) = d(F^* f), \quad F^*(d\omega) = d(F^*\omega)$$

and similarly for ∂ and $\bar{\partial}$.

Lemma 3.8 (Order Formula)

For a meromorphic 1-form ω on Y and $p \in X$:

$$\operatorname{ord}_p(F^*\omega) = (1 + \operatorname{ord}_{F(p)}(\omega)) \operatorname{mult}_p(F) - 1$$

Proof. Choose local coordinates w at p and z at $F(p)$ with $z = w^n$ where $n = \operatorname{mult}_p(F)$. If $\omega = (cz^k + \cdots)dz$, then:

$$F^*\omega = (cw^{nk} + \cdots)(nw^{n-1})dw = (cnw^{nk+n-1} + \cdots)dw$$

so $\operatorname{ord}_p(F^*\omega) = nk + n - 1$. □

The pullback of 2-forms is defined analogously by:

$$F^*(f\,dz \wedge d\bar{z}) = f(h(w), \overline{h(w)})|h'(w)|^2 dw \wedge d\bar{w}$$

These operations form the foundation for the calculus of differential forms on Riemann surfaces.

§3.3 Integration of forms

Definition. A **path** on a Riemann surface X is a continuous, piecewise C^∞ function $\gamma : [a, b] \to X$. The points $\gamma(a)$ and $\gamma(b)$ are the endpoints. A path is **closed** if $\gamma(a) = \gamma(b)$.

Paths can be reparametrized, reversed, and concatenated. Any path can be partitioned into finitely many C^∞ segments, each contained in a single chart domain.

Definition. Let ω be a C^∞ 1-form on X and γ a path. Choose a partition $\{\gamma_i\}$ such that each γ_i lies in a chart domain U_i with coordinate z_i. In each chart, write $\omega = f_i dz_i + g_i d\bar{z}_i$. Then define:

$$\int_\gamma \omega = \sum_i \int_{t_{i-1}}^{t_i} \left[f_i(z_i(t), \bar{z}_i(t)) z_i'(t) + g_i(z_i(t), \bar{z}_i(t)) \bar{z}_i'(t) \right] dt$$

This definition is independent of the choice of partition and charts due to the correct transformation properties of 1-forms.

Lemma 3.9 (Properties of Path Integration)

Let ω, ω_1, ω_2 be C^∞ 1-forms and γ a path.

- $\int_\gamma \omega$ is independent of parametrization
- $\int_\gamma (\lambda\omega_1 + \mu\omega_2) = \lambda \int_\gamma \omega_1 + \mu \int_\gamma \omega_2$
- If f is C^∞ near γ, then $\int_\gamma df = f(\gamma(b)) - f(\gamma(a))$
- $\int_{-\gamma} \omega = -\int_\gamma \omega$
- If $F : X \to Y$ is holomorphic, then $\int_{F_*\gamma} \omega = \int_\gamma F^*\omega$

Chains and Formal Sums

Definition. A **chain** on X is a finite formal sum $\gamma = \sum n_i \gamma_i$ where $n_i \in \mathbb{Z}$ and each γ_i is a path. The integral extends linearly:

$$\int_\gamma \omega = \sum n_i \int_{\gamma_i} \omega$$

Residues of Meromorphic 1-Forms

Definition. Let ω be a meromorphic 1-form with local expansion $\omega = (\sum_{n=-M}^{\infty} c_n z^n) dz$ near p. The **residue** of ω at p is $\mathrm{Res}_p(\omega) = c_{-1}$.

Lemma 3.10

For any small path γ enclosing p and no other poles:

$$\mathrm{Res}_p(\omega) = \frac{1}{2\pi i} \int_\gamma \omega$$

Proof. Choose a chart centered at p where $\omega = f(z)dz$. Then $\phi \circ \gamma$ is a small loop around 0 in $\mathbb{C}$, and by the classical residue theorem:

$$\int_\gamma \omega = \int_{\phi\circ\gamma} f(z)dz = 2\pi i c_{-1} = 2\pi i \operatorname{Res}_p(\omega)$$

□

Lemma 3.11

If f is meromorphic at p, then $\operatorname{Res}_p(df/f) = \operatorname{ord}_p(f)$.

Proof. In local coordinates, if $f(z) = cz^n + \cdots$ with $c \neq 0$, then:

$$\frac{df}{f} = \frac{ncz^{n-1} + \cdots}{cz^n + \cdots}dz = \left(\frac{n}{z} + \cdots\right)dz$$

so the residue is $n = \operatorname{ord}_p(f)$. □

Integration of 2-Forms and Stokes' Theorem

Definition. Let $D \subseteq X$ be a triangulable closed set and η a C^∞ 2-form. Triangulate D so each triangle T_i lies in some chart where $\eta = f_i dz \wedge d\bar{z}$. Then define:

$$\iint_D \eta = \sum_i \iint_{\phi_i(T_i)} f_i dz \wedge d\bar{z}$$

This is well-defined since the transformation rule for 2-forms gives $f(w)|T'(w)|^2$ under coordinate change $z = T(w)$, and $dz \wedge d\bar{z} = |T'(w)|^2 dw \wedge d\bar{w}$.

Definition. For a triangle T, let ∂T be the path traversing its boundary counterclockwise. For a triangulable set $D = \bigcup T_i$, define the **boundary chain** $\partial D = \sum \partial T_i$.

Theorem 3.1 (Stokes' Theorem)

For any triangulable closed set $D \subseteq X$ and C^∞ 1-form ω:

$$\int_{\partial D} \omega = \iint_D d\omega$$

Proof. By additivity, reduce to the case where D is a triangle contained in a single chart. Transfer to $\mathbb{C}$ via the chart map, where the result becomes Green's theorem:

$$\int_{\partial D} (f dx + g dy) = \iint_D \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right) dx dy$$

In complex notation, $d\omega = (\partial g/\partial z - \partial f/\partial \bar{z})dz \wedge d\bar{z}$, and the result follows. □

Theorem 3.2 (Residue Theorem)

Let ω be a meromorphic 1-form on a compact Riemann surface X. Then:

$$\sum_{p \in X} \operatorname{Res}_p(\omega) = 0$$

Proof. Let $p_1, \ldots, p_n$ be the poles of ω. For each p_i, choose a small path γ_i enclosing p_i but no other poles. Let U_i be the interior of γ_i and set $D = X - \bigcup U_i$. Then $\partial D = -\sum \gamma_i$ as chains, so:

$$\sum \mathrm{Res}_{p_i}(\omega) = \frac{1}{2\pi i} \sum \int_{\gamma_i} \omega = \frac{-1}{2\pi i} \int_{\partial D} \omega$$

By Stokes' theorem, $\int_{\partial D} \omega = \iint_D d\omega$. But $d\omega = 0$ on D since ω is holomorphic there, hence the result. □

Corollary 3.3. If f is a nonconstant meromorphic function on a compact Riemann surface X, then:

$$\sum_{p \in X} \mathrm{ord}_p(f) = 0$$

Proof. Apply the Residue Theorem to $\omega = df/f$, using $\mathrm{Res}_p(df/f) = \mathrm{ord}_p(f)$. □

Homotopy and Integration

Definition. Paths $\gamma_0, \gamma_1 : [a, b] \to X$ are **homotopic** if there exists a continuous map $\Gamma : [a, b] \times [0, 1] \to X$ such that $\Gamma(t, 0) = \gamma_0(t)$, $\Gamma(t, 1) = \gamma_1(t)$, and Γ is constant on $\{a\} \times [0, 1]$ and $\{b\} \times [0, 1]$.

Proposition 3.1

If γ_0 and γ_1 are homotopic and ω is closed ($d\omega = 0$), then:

$$\int_{\gamma_0} \omega = \int_{\gamma_1} \omega$$

Proof. Let D be the image of the homotopy (which is triangulable). Then $\partial D = \gamma_1 - \gamma_0$ up to reparametrization, so:

$$\int_{\gamma_1} \omega - \int_{\gamma_0} \omega = \int_{\partial D} \omega = \iint_D d\omega = 0$$

□

This shows that for closed 1-forms, the integral depends only on the homotopy class of the path. In particular, for holomorphic 1-forms, integration gives a well-defined group homomorphism from the fundamental group $\pi_1(X, p)$ to $\mathbb{C}$, which factors through the first homology group $H_1(X)$.